Expectation (Mean) of a Random Variable

Let X denote a **discrete** random variable with probability function $p(x)$ or if X is **continuous**, probability density function $f(x)$. Then the expected value of X, denoted $E(X)$ is defined to be:

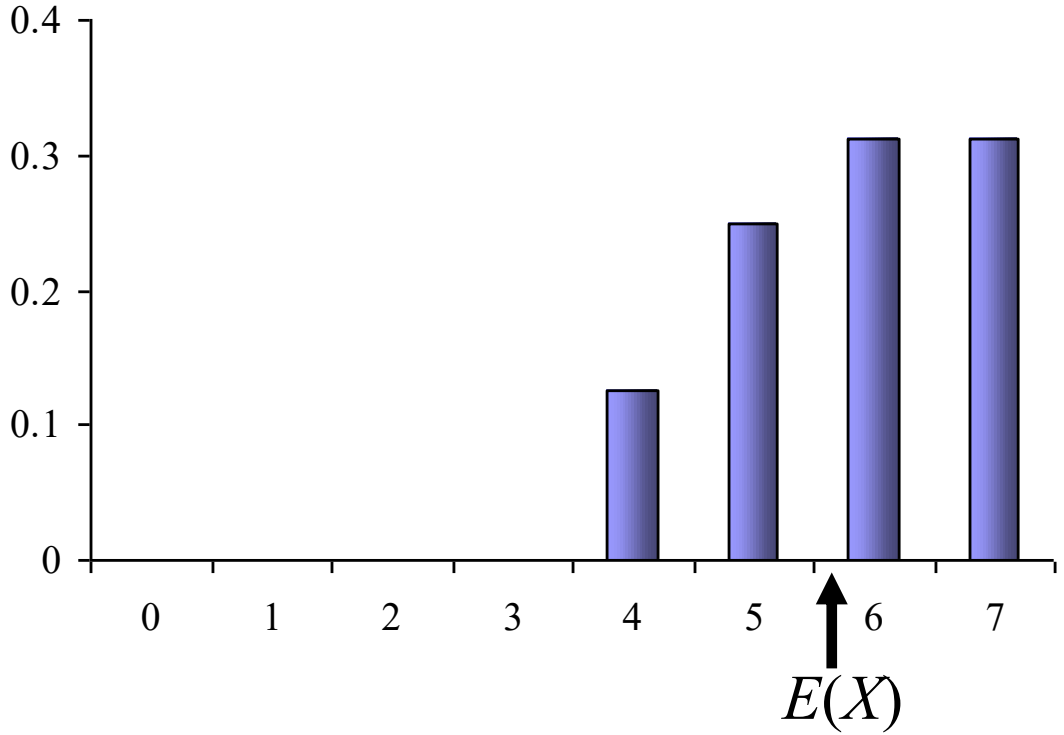
$$E(X) = \sum_x xp(x) = \sum_i x_i p(x_i)$$

And if X is **continuous** with probability density function $f(x)$, then

$$E(X) = \int_{-\infty}^{\infty} xf(x)dx$$

Interpretation of $E(X)$

1. The expected value of X, $E(X)$, is the center of gravity of the probability distribution of X.
2. The expected value of X, $E(X)$, is the **long-run average value** of X. (shown later –**Law of Large Numbers**)



Example 1:

A data scientist is modeling the monthly server outages for a cloud computing platform. The number of outages per month is represented by the random variable X. The probability distribution is known:

X (Outages)	0	1	2	3	4
$P(X)$	0.15	0.35	0.25	0.15	0.10

The company incurs a fixed maintenance cost of \$1,000 per month, plus an exponentially scaling penalty cost of \$500 per squared outage. The total monthly cost function is $C(X) = 500X^2 + 1000$.

a) Calculate the expected number of outages per month, $E[X]$.

b) Calculate the expected monthly cost, $E[C(X)]$.

Example 1:

X (Outages)	0	1	2	3	4
$P(X)$	0.15	0.35	0.25	0.15	0.10

Find $E(X)$.

Solution:

Multiply each outcome by its probability and sum the products.

$$E[X] = \sum x \cdot P(x)$$

$$E[X]$$
$$= (0)(0.15) + (1)(0.35) + (2)(0.25) + (3)(0.15) + (4)(0.10)$$
$$E[X] = 0 + 0.35 + 0.50 + 0.45 + 0.40$$
$$E[X] = 1.70 \text{ outages}$$

Exercise 1:

Find the expected value on a $1 bet on “red” in a European style roulette game (18 red numbers, 18 black and one green 0). How about on an American style roulette (18 red, 18 black and two green 0s)?

Example 2

Let the pdf of a continuous random variable be

$$f(x) = \frac{3}{2}(1 - x^2), \qquad 0 \leq x \leq 1$$

a) Find $E(X)$
b) Find the cdf $F(x)$

Example 2 Solution:

Let $f(x) = \frac{3}{2}(1 - x^2),\ 0 \le x \le 1$

Find $E(X)$

$$E(X) = \int_0^1 x f(x)\,dx = \int_0^1 x \frac{3}{2}(1 - x^2)\,dx$$

$$= \frac{3}{2}\int_0^1 (x - x^3)\,dx = \frac{3}{2}\left(\frac{x^2}{2}\Big|_0^1 - \frac{x^4}{4}\Big|_0^1 \right)$$

$$= \frac{3}{2}\left(\frac{1}{2} - 0 - \left(\frac{1}{4} - 0\right)\right) = \frac{3}{2}\left(\frac{1}{2} - \frac{1}{4}\right) = \frac{3}{8}$$

Example 3: Let $f(x) = a + bx^2,\ 0 \leq x \leq 1$

And $E(X) = 3/5$

Find a and b

Example: Let $f(x) = a + bx^2,\ 0 \le x \le 1$

And $E(X) = 3/5$

Find a and b

$$\int f(x)\,dx = 1 \quad \& \quad \int x f(x)\,dx = \frac{3}{5}$$

$$\int_0^1 (a+bx^2)\,dx = 1 \qquad \int_0^1 x(a+bx^2)\,dx = \frac{3}{5}$$

$$a x\Big]_0^1 + b\frac{x^3}{3}\Big]_{x=0}^1 = 1 \quad \& \quad a\frac{x^2}{2}\Big]_0^1 + b\frac{x^4}{4}\Big]_0^1 = \frac{3}{5}$$

$$a + \frac{b}{3} = 1 \qquad \frac{a}{2} + \frac{b}{4} = \frac{3}{5} \;\Big|\; 20$$

$$3a + b = 3 \qquad 10a + 5b = 12$$

$3a + b = 3 \Rightarrow b = 3 - 3a$

$10a + 5b = 12$

$10a + 5(3 - 3a) = 12$

$10a + 15 - 15a = 12$

$-5a = -3 \Rightarrow a = \frac{3}{5}$

$\Rightarrow b = 3 - 3\left(\frac{3}{5}\right) = \boxed{\frac{6}{5}}$

Exercise: Suppose we are observing a seven-game series where the teams are evenly matched, and the games are independent. Let X denote the length of the series. Find:

1. The distribution of X.
2. the expected value of X, $E(X)$.

Example: The Uniform distribution

Suppose X has a uniform distribution from a to b.

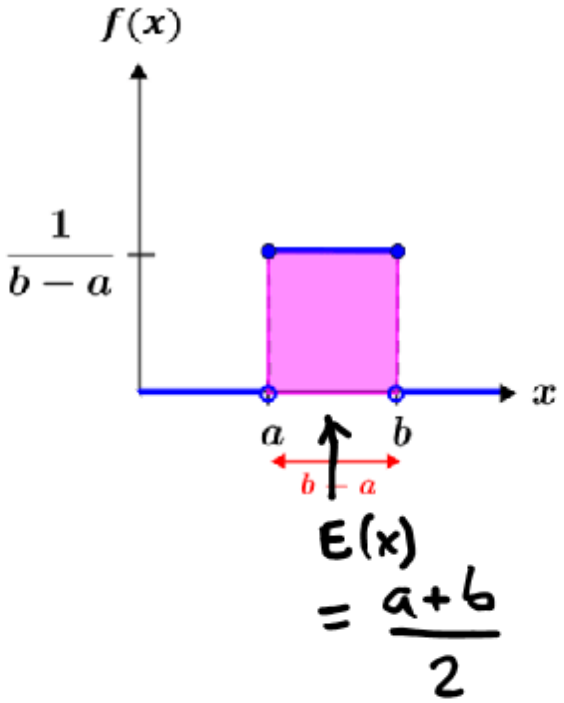
Then:

$$f(x) = \begin{cases} \dfrac{1}{b-a}, & a \le x \le b \\ 0, & x < a, x > b \end{cases}$$

The **expected value** of X is:

$$E(X) = \int_{-\infty}^{\infty} x f(x)\,dx = \int_{a}^{b} x \tfrac{1}{b-a}\,dx$$

$$= \left[\tfrac{1}{b-a} \frac{x^2}{2} \right]_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$



Expectation of Functions of Random Variables

Definition

Let X denote a **discrete** random variable with probability function $p(x)$ (probability density function $f(x)$ if X is **continuous**) then the expected value of $g(X)$, $E[g(X)]$ is defined to be:

$$E\left[g\left(X\right)\right]=\sum_{x} g\left(x\right)p\left(x\right)=\sum_{i} g\left(x_i\right)p\left(x_i\right)$$

and if X is **continuous** with probability density function $f(x)$

$$E\left[g\left(X\right)\right]=\int_{-\infty}^{\infty} g\left(x\right)f\left(x\right)dx$$

Example: Recall the example

X (Outages)	0	1	2	3	4
$P(X)$	0.15	0.35	0.25	0.15	0.10

Find $E[C(X)]$,

where $C(X) = 500X^2 + 1000$

Example: Recall the example

X (Outages)	0	1	2	3	4
$P(X)$	0.15	0.35	0.25	0.15	0.10

By the properties of expected value (see later slides), $E[aX^2 + b] = aE[X^2] + b$. Therefore, we first need to find $E[X^2]$, the expected value of the squared outages.

$$E[X^2] = \sum x^2 \cdot P(x)$$

$$= (0^2)(0.15) + (1^2)(0.35) + (2^2)(0.25) + (3^2)(0.15) + (4^2)(0.10) = 0 + 0.35 + 1.00 + 1.35 + 1.60 = 4.30$$

Now, plug $E[X^2]$ into the expected cost function:

$$E[C(X)] = E[500X^2 + 1000] = 500E[X^2] + 1000$$

$$E[C(X)] = 500(4.30) + 1000$$

$$E[C(X)] = 2150 + 1000 = \$3{,}150$$

Example

Let X be a continuous r.v. with pdf

$$f(x) = 3x^2, \qquad x \in (0,1)$$

Find $\mathrm{E}\left(\frac{1}{X}\right)$

Exercise:

Let X be a continuous r.v. with pdf

$$f(x) = x^2\left(2x + \frac{3}{2}\right), \qquad x \in (0,1)$$

Find $\mathrm{E}\left(\frac{2}{X} + 3\right)$

Moments of Random Variables

Definition

Let X be a random variable (discrete or continuous), then the kth **moment** of X is defined to be:

$$\mu_k = E\left(X^k\right)$$

$$= \begin{cases} \sum_x x^k p(x) & \text{if } X \text{ is discrete} \\ \int_{-\infty}^{\infty} x^k f(x)\,dx & \text{if } X \text{ is continuous} \end{cases}$$

The first moment of X, $\mu = \mu_1 = E(X)$ is the center of gravity of the distribution of X. The higher moments give different information regarding the distribution of X.

Definition: Let X be a random variable. Then the kth *central moment* of X is defined to be:

$$\mu_k^0 = E[(X - \mu)^k]$$

Where $\mu = \mu_1 = E(X)$ is the first moment of X.

Notes:

- Moments of a function are related to its shape.
- If the function represents mass density, then the zeroth moment is the total mass, the first moment is the center of mass, and the second moment is the moment of inertia.
- If the function is a probability distribution, then the first moment is the expected value, the second central moment is the variance, the third standardized moment is the skewness, and the fourth standardized moment is the kurtosis.

The central moments describe how the probability distribution is distributed about its center of gravity, μ:

$$\mu_1^0 = E(X - \mu) = 0$$

$$\mu_2^0 = E\left[(X-\mu)^2\right] \quad = 2^{nd} \text{ central moment.}$$

depends on the *spread* of the probability distribution of X about μ.

$\mu_2^0 = E\left[(X-\mu)^2\right]$ is called the variance of X.

and is denoted by the symbol var(X).

The **population standard deviation** (σ), defined scientifically in probability theory and statistics as the principal square root of the second central moment (μ_2^0) of a random variable X:

$$\sigma = \sqrt{\mu_2^0} = \sqrt{\mathbb{E}[(X-\mu)^2]}$$

Then

$$\boxed{\operatorname{var}(X) = \mu_2^0 = E\left[\left(X-\mu\right)^2\right] = \sigma^2}$$

Relevance in Data Science

Feature Scaling (Z-score Normalization): Used to transform data to zero mean and unit variance ($z = \frac{x-\mu}{\sigma}$), preventing features with larger scales from dominating optimization algorithms such as Gradient Descent, Support Vector Machines (SVM), and k-Nearest Neighbors (k-NN).

Spread & Anomaly Detection: Serves as a dispersion metric to define threshold bounds (e.g., 3σ rule) for identifying statistical outliers and measuring noise in data pipelines.

Model Evaluation & Risk Assessment: Measures prediction error dispersion, variance in model performance across cross-validation folds, and volatility in time-series models.

The **third central moment** (μ_3^0) is a statistical measure that quantifies the degree of asymmetry in a random variable's probability distribution relative to its mean.

$$\mu_3^0 = \mathbb{E}[(X - \mu)^3]$$

Relevance in Data Science

Distribution Asymmetry Identification: Determines whether data points trail off toward higher positive values ($\mu_3^0 > 0$, right/positive skew) or lower negative values ($\mu_3^0 < 0$, left/negative skew).

Feature Engineering & Data Transformations: Drives decisions on applying power transformations (such as Log, Square Root, or Box-Cox) to normalize feature distributions before training parametric models sensitive to skewness (e.g., Linear Regression, Linear Discriminant Analysis).

Tail Risk & Anomaly Detection: Cubing deviations gives disproportionate weight to extreme tail events, making it critical in financial data analysis, risk management, and detecting asymmetric outlier clusters.

Model Diagnostic Validation: Used in goodness-of-fit testing (such as the Jarque-Bera test) to verify whether residual errors or dataset features violate the assumption of multivariate normality.

The third central moment

$$\mu_3^0 = E\left[\left(X-\mu\right)^3\right]$$

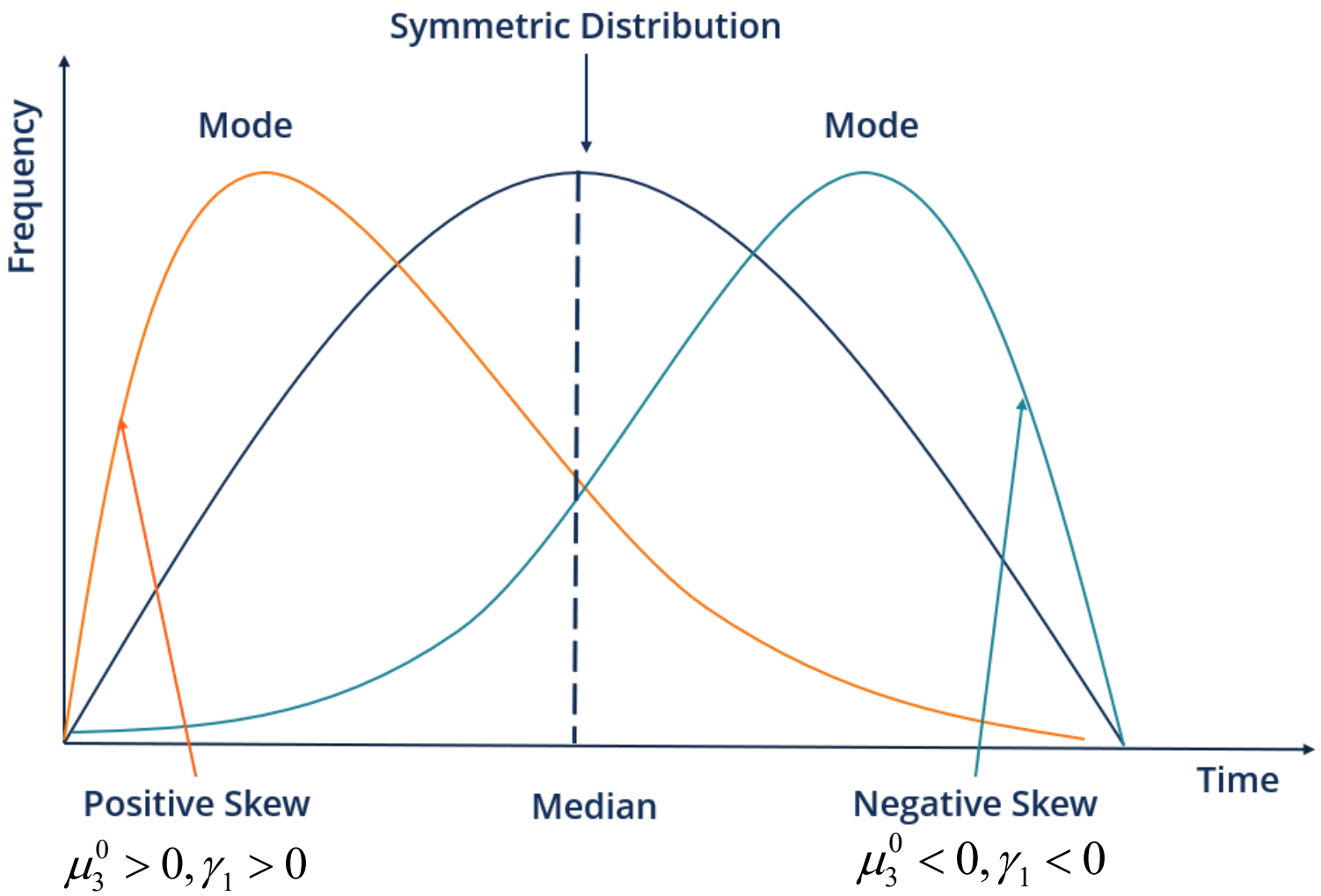
contains information about the **skewness** of a distribution.

Relationship to Skewness (γ_1): Normalizing the third central moment by the standard deviation cubed (σ^3) yields **skewness**, a scale-free measure of distribution symmetry:

$$\gamma_1 = \frac{\mu_3^0}{\sigma^3} = \frac{\mathbb{E}[(X-\mu)^3]}{(\mathbb{E}[(X-\mu)^2])^{3/2}}$$

Skewness

$$\mu_3^0 = 0, \gamma_1 = 0$$



The **fourth central moment** (μ_4^0) is a statistical measure that quantifies the tail weight and propensity for extreme outcomes in a random variable's probability distribution relative to its mean.

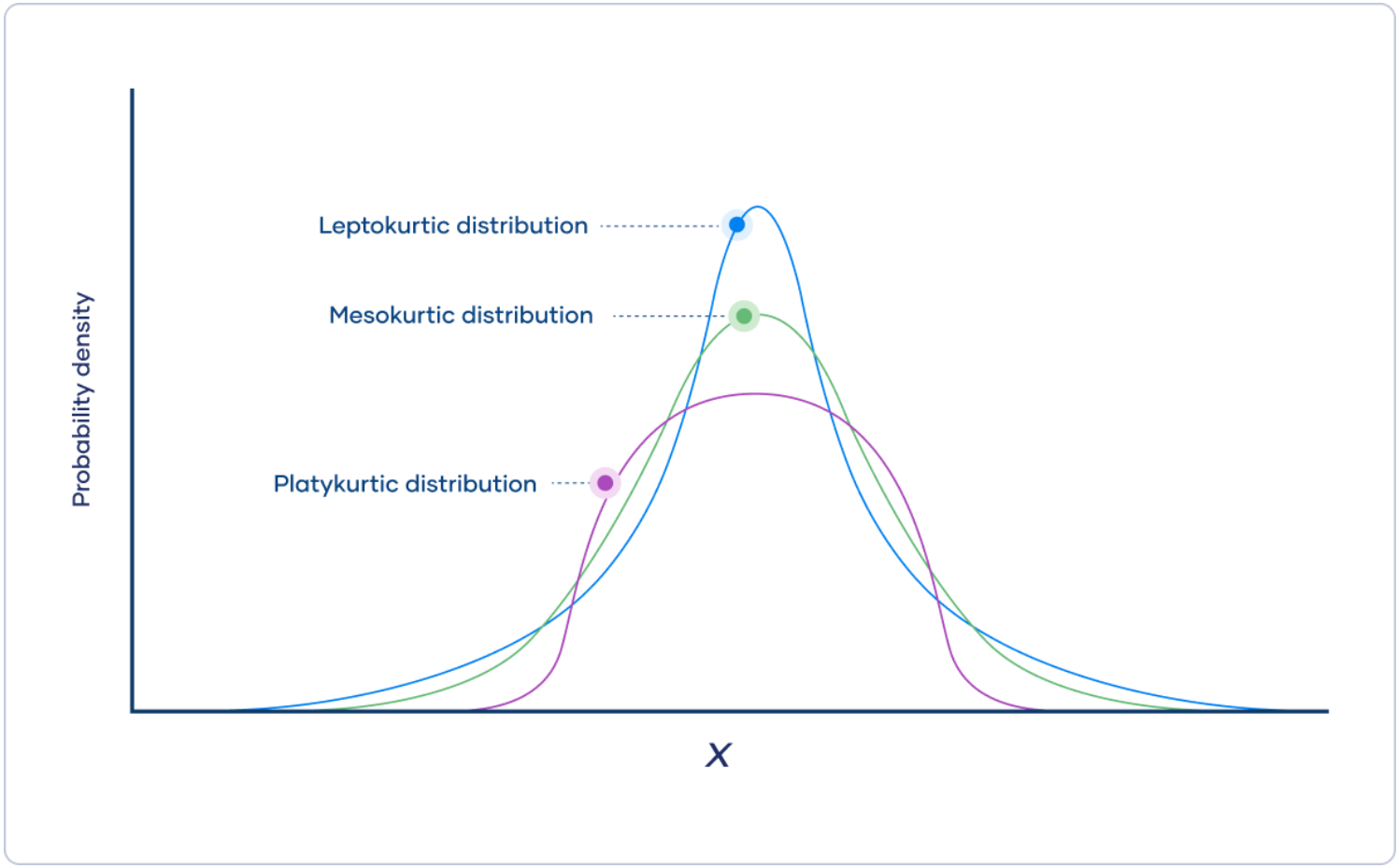
$$\mu_4^0 = \mathbb{E}[(X - \mu)^4]$$

Also contains information about the **shape** of a distribution. The property of shape that is measured by the fourth central moment is called **kurtosis**

Relationship to Kurtosis (Kurt$[X]$**):** Normalizing the fourth central moment by the square of the variance (σ^4) yields **kurtosis**, a scale-invariant measure of distribution tail-heaviness:

$$\text{Kurt}[X] = \frac{\mu_4^0}{\sigma^4} - 3 = \frac{\mathbb{E}[(X - \mu)^4]}{(\mathbb{E}[(X - \mu)^2])^2} - 3$$

Types of kurtosis



Kurtosis is **a measure of whether the data are heavy-tailed or light-tailed relative to a normal distribution**. That is, data sets with high kurtosis tend to have heavy tails, or outliers. Data sets with low kurtosis tend to have light tails, or lack of outliers.

How about higher moments?

Higher-order moments like the 5th (hyperskewness) and 6th (hyperkurtosis) do have applications in data science, but they are highly specialized.

In standard machine learning, practitioners rarely venture beyond the 4th moment because raising deviations to the 5th or 6th power makes the metric extraordinarily sensitive to outliers. However, in specific domains where modeling extreme non-Gaussian behavior is the entire goal, higher-order moments become useful.

Here are the primary data science and analytical applications for moments beyond the 4th:

1. Quantitative Finance and Extreme Risk Modeling

Options Pricing: Financial models often use Gram-Charlier or Edgeworth expansions to adjust the Black-Scholes formula for non-normal market behaviors. These expansions rely on higher-order moments to price exotic options by accounting for the exact shape of extreme market crashes or windfalls.

2. Signal Processing and Computer Vision

Texture Analysis: In image processing and computer vision, higher-order statistical moments are extracted from pixel intensity histograms as features. They help machine learning models classify complex micro-textures (like medical tissue scans or satellite imagery terrain) where lower moments fail to capture the subtle, high-frequency noise patterns.

3. Physics and Astronomy

Data scientists working in astrophysics use higher-order moments to study the large-scale structure of the universe. For example, analyzing the 5th and 6th moments of matter distribution helps detect non-Gaussianity in cosmic microwave background radiation, which tests theories of quantum fluctuations in the early universe.

Example: Let X be a discrete r.v. with $p(x)$

X	1	2	3
$p(x)$	½	1/3	1/6

Find:

a) E(X)

b) Standard deviation σ

c) Third moment μ_3

d) Third central moment μ^0_3

Example: Let X be a discrete r.v. with $p(x)$

X	1	2	3
$p(x)$	½	1/3	1/6

Find:

a) $E(X) = 1 * \left(\frac{1}{2}\right) + 2 * \left(\frac{1}{3}\right) + 3 * \left(\frac{1}{6}\right) = \frac{5}{3} = \mu$

b) Standard deviation σ

$$Var(X) = \sum (x-\mu)^2 p(x)$$

$$= \left(1-\frac{5}{3}\right)^2\left(\frac{1}{2}\right) + \left(2-\frac{5}{3}\right)^2\left(\frac{1}{3}\right) + \left(3-\frac{5}{3}\right)^2\left(\frac{1}{6}\right) = \frac{5}{9}$$

$$\Rightarrow \sigma = \sqrt{\frac{5}{9}} = \frac{\sqrt{5}}{3}$$

c) $\mu_3 = \sum x^3 \cdot p(x) = 1^3\left(\frac{1}{2}\right) + 2^3\left(\frac{1}{3}\right) + 3^3\left(\frac{1}{6}\right) = \frac{23}{3}$

d) ... $\frac{7}{27}$

Example: The uniform distribution from 0 to 1

$$f(x)=\begin{cases}1 & 0\le x\le 1\\ 0 & x<0, x>1\end{cases}$$

Finding the moments

$$\mu_k=\int_{-\infty}^{\infty}x^k f(x)\,dx=\int_0^1 x^k 1\,dx=\left[\frac{x^{k+1}}{k+1}\right]_0^1=\frac{1}{k+1}$$

Finding the central moments:

$$\mu_k^0 = \int_{-\infty}^{\infty} (x-\mu)^k f(x)\,dx = \int_0^1 \left(x-\tfrac{1}{2}\right)^k 1\,dx$$

making the substitution $w = x - \frac{1}{2}$

$$\mu_k^0 = \int_{-\frac{1}{2}}^{\frac{1}{2}} w^k\,dw = \left[\frac{w^{k+1}}{k+1}\right]_{-\frac{1}{2}}^{\frac{1}{2}} = \frac{\left(\frac{1}{2}\right)^{k+1} - \left(-\frac{1}{2}\right)^{k+1}}{k+1}$$

$$= \frac{1-(-1)^{k+1}}{2^{k+1}(k+1)} = \begin{cases} \dfrac{1}{2^k(k+1)} & \text{if } k \text{ even} \\ 0 & \text{if } k \text{ odd} \end{cases}$$

Hence $\mu_2^0 = \frac{1}{2^2(3)} = \frac{1}{12}, \mu_3^0 = 0, \mu_4^0 = \frac{1}{2^4(5)} = \frac{1}{80}$

Thus

$$\text{var}(X) = \mu_2^0 = \frac{1}{12}$$

The standard deviation

$$\sigma = \sqrt{\text{var}(X)} = \sqrt{\mu_2^0} = \frac{1}{\sqrt{12}}$$

The measure of skewness

$$\gamma_1 = \frac{\mu_3^0}{\sigma^3} = 0$$

The measure of kurtosis

$$\gamma_1 = \frac{\mu_4^0}{\sigma^4} - 3 = \frac{1/80}{(1/12)^2} - 3 = -1.2$$

Properties of Expected Value

Rules:

1. $E[c] = c$ where c is a constant

Proof

if $g(X) \equiv c$ then $E[g(X)] = E[c] = \int_{-\infty}^{\infty} cf(x)\,dx$

$$= c\int_{-\infty}^{\infty} f(x)\,dx = c$$

The proof for discrete random variables is similar.

2. $E[aX+b]=aE[X]+b$ where a,b are constants

Proof

if $g(X)\equiv aX+b$ then $E[aX+b]=\int_{-\infty}^{\infty}(ax+b)f(x)dx$

$$=a\int_{-\infty}^{\infty}xf(x)dx+b\int_{-\infty}^{\infty}f(x)dx$$

$$=aE(X)+b$$

The proof for discrete random variables is similar.

3. $\operatorname{var}(X) = \mu_2^0 = E\left[(X-\mu)^2\right]$

$$= E\left(X^2\right) - \left[E(X)\right]^2 = \mu_2 - \mu_1^2$$

Proof

$$\operatorname{var}(X) = E\left[(X-\mu)^2\right] = \int_{-\infty}^{\infty} (x-\mu)^2 f(x)\,dx$$

$$= \int_{-\infty}^{\infty} \left(x^2 - 2x\mu + \mu^2\right) f(x)\,dx$$

$$= \int_{-\infty}^{\infty} x^2 f(x)\,dx - 2\mu \int_{-\infty}^{\infty} x f(x)\,dx + \mu^2 \int_{-\infty}^{\infty} f(x)\,dx$$

$$= E\left(X^2\right) - 2\mu\left[E(X)\right] + \mu^2 = \mu_2 - \mu^2 = \mu_2 - \mu_1^2$$

The proof for discrete random variables is similar.

4. $\operatorname{var}(aX+b) = a^2 \operatorname{var}(X)$

Proof

$$\mu_{aX+b} = E[aX+b] = aE[X]+b = a\mu+b$$

$$\operatorname{var}(aX+b) = E\left[(aX+b-\mu_{aX+b})^2\right]$$

$$= E\left[(aX+b-[a\mu+b])^2\right]$$

$$= E\left[a^2(X-\mu)^2\right]$$

$$= a^2 E\left[(X-\mu)^2\right] = a^2 \operatorname{var}(X)$$

Example:

Let X be a continuous r.v. with pdf

$$f(x) = x^2\left(2x + \frac{3}{2}\right), \qquad x \in (0,1)$$

Find $\mathrm{Var}\left(\frac{2}{X} + 3\right)$

$$\mathrm{Var}\left(\frac{2}{X}+3\right) = \mathrm{Var}\left(\underset{a}{2}\cdot\frac{1}{X}+\underset{b}{3}\right) = \underset{b}{2}^2\,\mathrm{Var}\left(\frac{1}{X}\right) \quad \text{Rule \#4}$$

$$\mathrm{Var}\left(\frac{1}{X}\right) \overset{\text{Rule \#3}}{=} E\left(\left(\frac{1}{X}\right)^2\right) - \left[E\left(\frac{1}{X}\right)\right]^2$$

$$= \int_0^1 \frac{1}{x^2}\cdot x^2\left(2x+\frac{3}{2}\right)dx - \left[\int_0^1 \frac{1}{x}\cdot x^2\left(2x+\frac{3}{2}\right)dx\right]^2$$

$$= \int_0^1 \left(2x+\frac{3}{2}\right)dx - \left[\int_0^1 x\left(2x+\frac{3}{2}\right)dx\right]^2 = \frac{71}{144}$$

Moment generating functions

Recall

$$E\left[g(X)\right]=\begin{cases}\sum_{x} g(x)\,p(x) & \text{if } X \text{ is discrete}\\ \int_{-\infty}^{\infty} g(x)\,f(x)\,dx & \text{if } X \text{ is continuous}\end{cases}$$

Definition

Let X denote a random variable, Then the moment generating function of X, $m_X(t)$ is defined by:

$$m_X(t)=E\left[e^{tX}\right]=\begin{cases}\sum_{x} e^{tx}\,p(x) & \text{if } X \text{ is discrete}\\ \int_{-\infty}^{\infty} e^{tx}\,f(x)\,dx & \text{if } X \text{ is continuous}\end{cases}$$

Example

The Binomial distribution $X \sim \text{Bin}(n, p)$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0, 1, 2, \text{K}\ , n$$

The moment generating function of X, $m_X(t)$ is:

$$M_X(t) = (pe^t + q)^n$$

Example

The Binomial distribution $X \sim \text{Bin}(n, p)$

The Moment Generating Function $M_X(t)$ of a discrete random variable is the expected value of e^{tX}: $M_X(t) = \mathbb{E}[e^{tX}]$

To derive the MGF, substitute the PMF into the expectation formula and sum over all possible values of x:

$$M_X(t) = \sum_{x=0}^{n} e^{tx} \binom{n}{x} p^x q^{n-x}$$

Use exponent rules to combine e^{tx} and p^x:

$$M_X(t) = \sum_{x=0}^{n} \binom{n}{x} (pe^t)^x q^{n-x}$$

This expression perfectly matches the algebraic structure of the Binomial Theorem, which states that $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^k b^{n-k}$.

By substituting $a = pe^t$ and $b = q$, the sum simplifies entirely:

$$M_X(t) = (pe^t + q)^n$$

Example

The Binomial distribution $X \sim \text{Bin}(n, p)$

$$p(x) = \binom{n}{x} p^x (1-p)^{n-x} \qquad x = 0,1,2,\text{K}\ ,n$$

The moment generating function of X, $m_X(t)$ is:

$$M_X(t) = (pe^t + q)^n$$

Why do we care?

$$E(X) = M_X'(0)$$

In our case, $M_X'(t) = n(pe^t + q)^{n-1}(pe^t)$

$$\Rightarrow M_X'(0) = n(pe^0 + q)^{n-1}(pe^0)$$

$$= n(p * 1 + q)^{n-1}(p) = np(p+q)^{n-1} = np(1)^{n-1} = np$$

3. MGF of uniform distribution

Let $X \sim \mathrm{U}(0, 1)$, that is, $f(x) = 1$, $x \in [0, 1]$.

Derive mgf:

$$m(t) = \int_0^1 e^{xt} \cdot 1 dx = \frac{1}{t} \int_0^1 e^{xt}\, d(xt)$$

$$= \frac{1}{t} e^{xt} \Big]_{x=0}^{1} = \frac{e^t - 1}{t}, t \neq 0$$

Notice that $m(0) = 1$ must be calculated separately.

Properties of
Moment Generating Functions

1. $m_X(0) = 1$

$m_X(t) = E\left(e^{tX}\right)$, hence $m_X(0) = E\left(e^{0\cdot X}\right) = E(1) = 1$

2. $$m_X(t) = 1 + \mu_1 t + \frac{\mu_2}{2!}t^2 + \frac{\mu_3}{3!}t^3 + \mathrm{K} + \frac{\mu_k}{k!}t^k + \mathrm{K}$$

3. $$m_X^{(k)}(0) = \frac{d^k}{dt^k} m_X(t)\bigg|_{t=0} = \mu_k$$

MGF not always helpful ☹

Let $X \sim \mathrm{U}(0, 1)$, that is, $f(x) = 1$, $x \in [0, 1]$.

We know $E(X) = \frac{0+1}{2} = \frac{1}{2}$, but we will derive this with the mgf:

$$m(t) = \frac{e^t - 1}{t}, t \neq 0$$

Notice that $m(0) = 1$.

The expected value will be a limit of the derivative:

$$E(X) = m'(0) = \left(\frac{\mathrm{e}^t - 1}{t}\right)' = \frac{te^t - e^t + 1}{t^2} \to \frac{1}{2}$$

as $t \to 0$.

Also note, that you need to use L'Hôpital's rule!